

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Smart Campus

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Abhishek Kumar

2210991167

Supervised By:

Dr. Shikha Tuteja

Associate Professor

Chitkara University



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY

CONTENTS

S.No.	Title	Page No.
	Acknowledgement	iv
	List of Figures and Tables	v
	Abstract	
1	Introduction	1
2	Methodology	
3	Tools and Technologies	
4	Implementation	
5	Major Findings/Outcomes/Output/Results	
6	Conclusion and Future Scope	
	References	
	Appendices	

DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Smart Campus**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Shikha Tuteja. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date: 13 May 2026

(Name of the Student)

Abhishek Kumar

2210991167

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Shikha Tuteja

Associate Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my mentor Dr. Shikha Tuteja mam for their continuous support and guidance throughout this project. Their insights and suggestions helped me improve the system effectively.

I also thank Chitkara University for providing the necessary resources and environment to complete this project successfully.

Abstract

The Smart Campus Booking System is a full-stack web application developed to streamline campus resource management and event coordination. The system integrates real-time seat availability tracking and role-based access control to ensure efficient utilization of campus facilities.

The platform offers functionalities such as resource booking, event creation with seat selection, and user management for students, faculty, and administrators. It also includes booking status tracking, participant management, and administrative control features.

By combining modern web technologies with structured resource management, the Smart Campus Booking System aims to improve transparency, reduce manual effort, and enhance overall efficiency in campus operations. The system provides an accessible and reliable solution for managing institutional resources and events.

Introduction

Modern educational institutions require efficient systems to manage campus resources such as lecture halls, seminar halls, libraries, and event spaces. Traditional booking methods often involve manual processes, leading to scheduling conflicts, lack of transparency, and inefficient utilization of resources. As institutions grow, the need for a centralized and automated system becomes increasingly important.

The Smart Campus Booking System is designed to address these challenges by providing a digital platform for booking and managing campus resources. The system enables users to check real-time availability, select seats, and manage bookings efficiently.

In addition to resource booking, the platform supports event creation and participation with seat selection capabilities. It also includes role-based access for students, faculty, and administrators, ensuring controlled and secure operations. By integrating modern technologies with practical functionality, the system aims to improve efficiency, organization, and user experience within campus environments.

Methodology

The Smart Campus Booking System was developed using a structured full-stack approach based on the MERN architecture. The process started with requirement analysis, where issues in traditional manual booking systems such as scheduling conflicts and lack of transparency were identified.

The system was designed using a client-server model, where the frontend and backend communicate through RESTful APIs. React.js was used to build a responsive user interface, while Node.js and Express.js handled server-side logic. MongoDB was used for efficient data storage.

The development followed a modular approach, dividing the system into authentication, booking, event management, and admin modules. Each module was implemented and tested individually to ensure proper functionality.

Security measures such as JWT-based authentication and password encryption were applied to protect user data. Finally, the system was tested to ensure smooth performance, reliability, and scalability.

Tools and Technologies

Backend:

- Node.js: JavaScript runtime used for server-side development
- Express.js: Web framework for building RESTful APIs
- MongoDB: NoSQL database for storing application data
- Mongoose: Object Data Modeling library for MongoDB
- JSON Web Token (JWT): Used for authentication and secure session management
- bcryptjs: Library for hashing user passwords

Frontend:

- React: JavaScript library for building interactive user interfaces
- React Router DOM: Handles client-side routing
- TailwindCSS: Utility-first CSS framework for styling
- Axios: HTTP client for API communication

Implementation

1. User Registration and Login:

New users register by providing basic details such as name, email, and password. The password is securely hashed before being stored in the database. Existing users can log in using their credentials and receive a JSON Web Token (JWT) for authenticated access.

2. Dashboard Access:

After successful login, users are directed to the dashboard, which displays key information such as booking history, event participation, and quick access to system features. The system provides real-time updates on bookings and activities.

3. Resource Selection:

Users can browse and select available campus resources such as canteen, seminar hall, lecture hall, library, and other facilities. Each resource includes details such as capacity, availability, and booking options.

4. Seat Selection Process:

When a user selects a resource, the system displays a seat map showing available and booked seats. Users can select their preferred seats based on real-time availability.

5. Booking Submission:

After selecting seats, users provide details such as date, time, and purpose of booking. The booking request is then submitted to the system and stored in the database.

6. Booking Processing and Status:

The system assigns a status to each booking such as pending, approved, or rejected. Administrators can review and update booking status accordingly.

7. Event Creation and Participation:

Users can create events by providing details such as event name, description, date, and capacity. A unique room code is generated for each event. Other users can join events using the room code and select seats.

RESULTS

- The Smart Campus Booking System has been successfully developed and tested, demonstrating reliable performance across all core functionalities. The system effectively manages campus resources and event coordination with real-time seat availability and booking operations. The seat selection interface ensures accurate allocation and prevents booking conflicts.
- The resource booking module functions smoothly, allowing users to book facilities based on availability and requirements. The event management feature enables users to create and join events efficiently using unique room codes and seat selection.
- The booking management system accurately records booking data, including resource type, date, time, and seat allocation. It also maintains booking status such as pending, approved, and rejected, enabling proper monitoring and control.
- The admin panel provides effective management of users, bookings, and events, ensuring smooth system operation. Overall, the system delivers a responsive user experience with efficient data handling and real-time updates. The Smart Campus Booking System effectively improves resource utilization and simplifies campus management processes.

Conclusion

- The Smart Campus Booking System successfully addresses the challenges of managing campus resources and event coordination within educational institutions. By integrating real-time seat availability, resource booking, and event management, the system provides an efficient and practical solution for organizing campus activities.
- The platform ensures that users can book resources without conflicts through realtime seat selection and structured booking processes. Its organized approach, combined with role-based access and administrative control, helps maintain transparency and efficient utilization of resources.
- Built using modern web technologies, the system is scalable, efficient, and easy to maintain. The Smart Campus Booking System demonstrates how technology can be effectively utilized to improve resource management and operational efficiency in institutional environments, making it a valuable solution for modern campuses.

Future Scope

- **Mobile Application:** Development of native mobile applications for Android and iOS to provide better accessibility and enable users to book resources and manage events on the go.
- **Advanced Analytics:** Implementation of detailed analytics with graphical insights to help administrators track resource usage, booking trends, and optimize utilization.
- **IoT Integration:** Integration with smart devices such as automated entry systems or smart displays to provide real-time updates on room availability and bookings.
- **Multi-language Support:** Adding support for multiple languages to make the platform accessible to a wider range of users across different regions.
- **Notification System:** Introducing real-time notifications for booking confirmations, status updates, and event reminders.
- **Institutional Integration:** Developing features to integrate with existing college management systems for seamless data sharing and automation.
- **Voice Assistance:** Incorporating voice commands for easier interaction while searching resources or booking facilities.
- **Custom Resource Management:** Allowing administrators to create and configure custom resources, seating layouts, and booking rules based on institutional needs.